Data Preprocessing, Feature Engineering & ML Pipeline

AI/ML Internship — Task 2 | Reproducible employee attrition data preparation

Project overview

This report documents real, computed preprocessing results for the IBM HR Analytics Employee Attrition sample. The work prepares a reusable pipeline for a later task; it does not train a classifier. Attrition stays separate from transformed features.

Dataset and data quality

Source sample: 1,470 rows × 35 fields. Missing cells: 0; exact duplicate rows: 0. After cleaning: 1,470 rows, 0 missing cells, 0 duplicates removed, and 0 missing-target rows removed. Text was trimmed and target labels normalized. Feature imputation remains in the train-fitted pipeline.

Target: No 83.88% / Yes 16.12%. Split: 1176 train / 294 test, stratified, random_state=42. No transformation is fitted on the test partition.

Feature engineering and pipeline

CareerStage and CompanyTenureBand are derived from Age and YearsAtCompany. EmployeeNumber is omitted as an identifier. The fitted Pipeline uses median numeric imputation, training-only IQR capping for continuous variables, StandardScaler, categorical mode imputation, and OneHotEncoder(handle_unknown='ignore'). Encoders, imputers, scaling, and IQR fences are learned from X_train only. The cleaned table retains all rows; outlier flags do not cause row deletion.

Outlier summary (1.5 × IQR)

Feature	Flagged	Share
TrainingTimesLastYear	238	16.2%
PerformanceRating	226	15.4%
MonthlyIncome	114	7.8%
YearsSinceLastPromotion	107	7.3%
YearsAtCompany	104	7.1%
StockOptionLevel	85	5.8%
TotalWorkingYears	63	4.3%
NumCompaniesWorked	52	3.5%

Observed findings and caveats

The sample has 16.12% records labelled Yes and 83.88% labelled No, indicating class imbalance. The tenure-band chart reports descriptive shares only. IQR flags identify unusual values, not errors; rows are retained. The dataset is a fictional, static sample and cannot support causal or population-wide conclusions. Correlation or group differences do not establish causation.

Charts

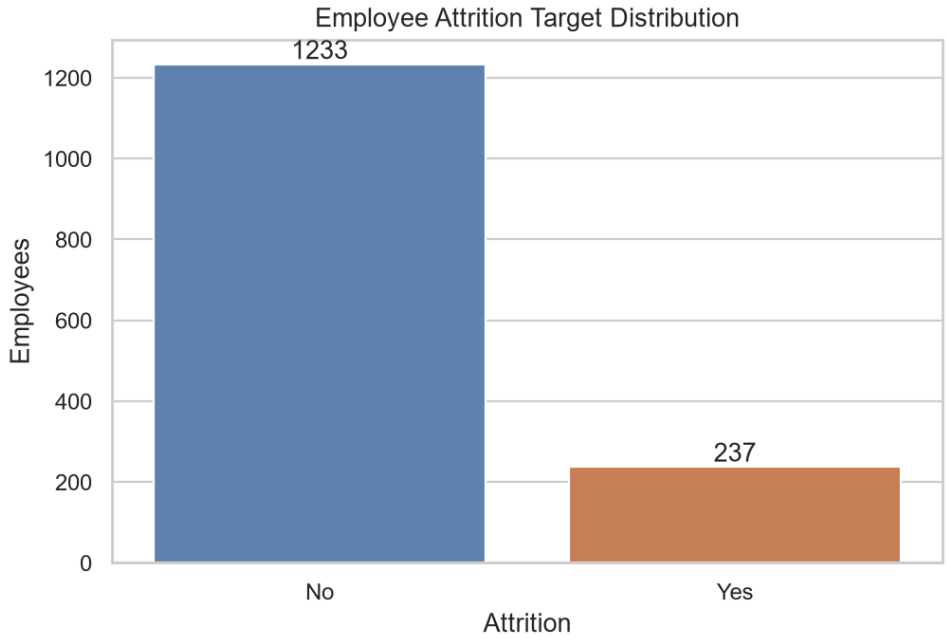


Figure 1. Counts of the Attrition target classes.

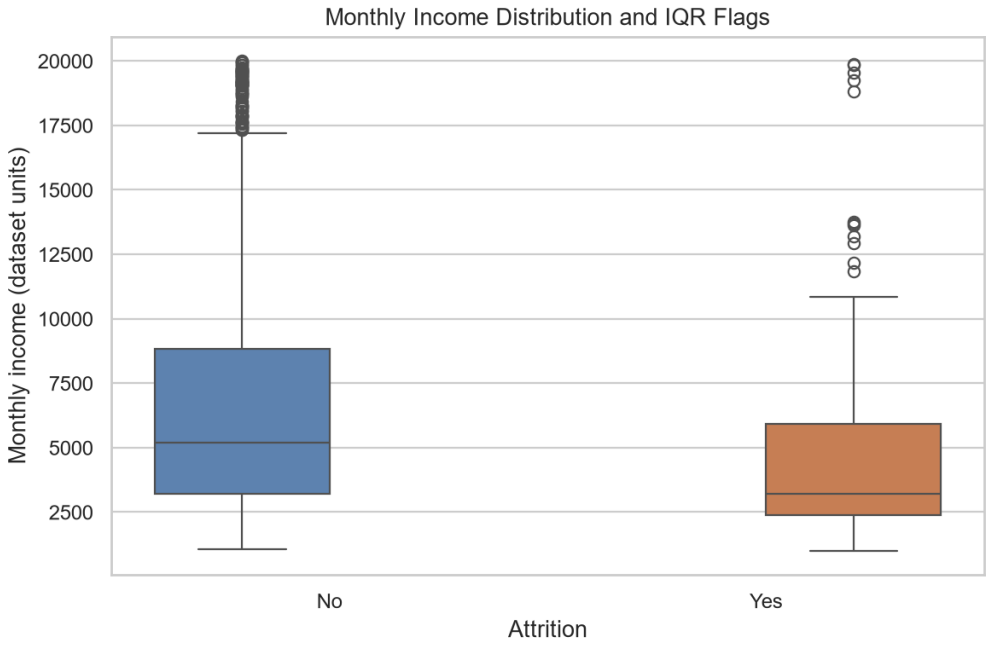


Figure 2. Monthly income distributions by target class; points beyond whiskers are potential outliers.

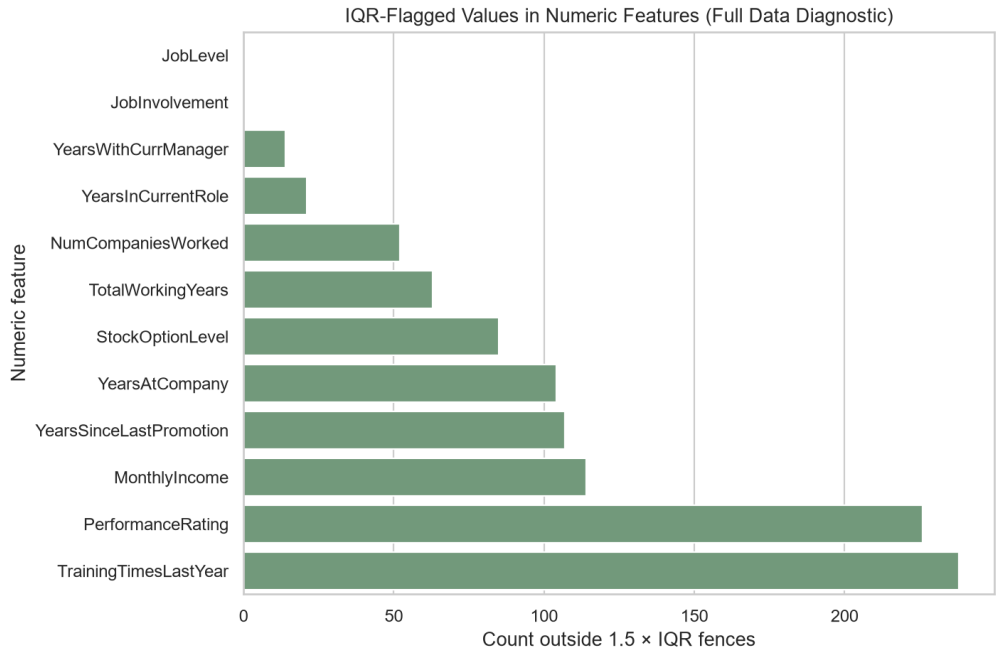


Figure 3. Numeric features with the highest 1.5×IQR flag counts.

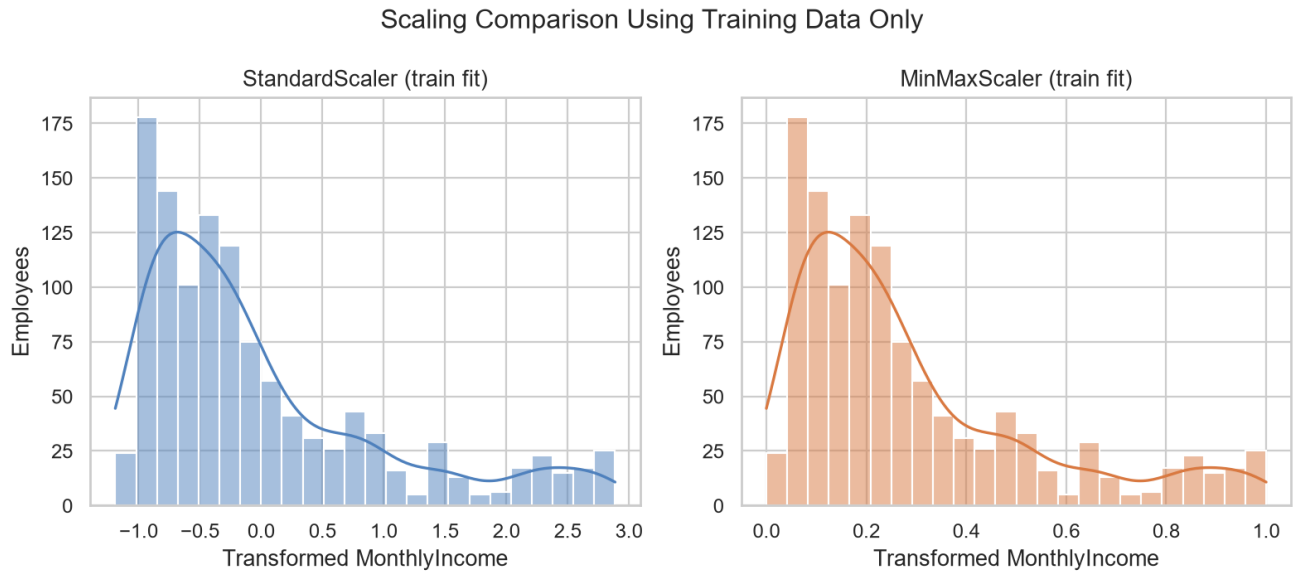


Figure 4. Training-only StandardScaler and MinMaxScaler transformations for MonthlyIncome.

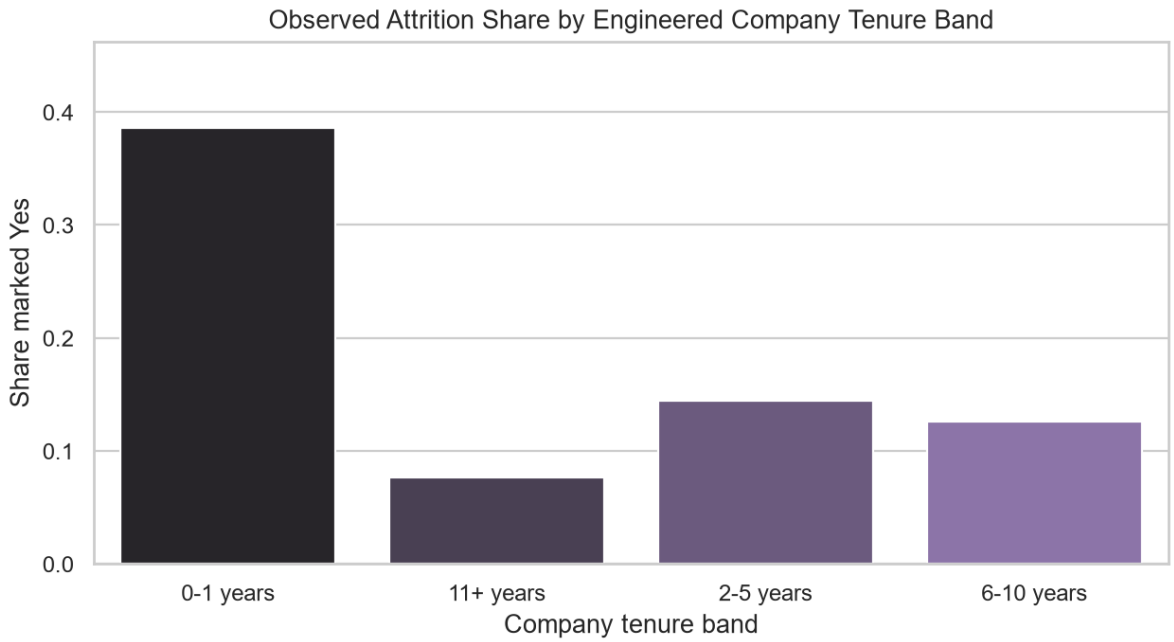


Figure 5. Observed attrition share by engineered company-tenure band.

Reproducibility, interpretation, and conclusion

The stratified split uses random_state=42 and test_size=0.2. The train matrix shape is [1176, 58] and test matrix shape is [294, 58]. LabelEncoder and OneHotEncoder are compared on the same nominal BusinessTravel feature; the reusable pipeline uses OneHotEncoder because integer label codes would imply a false order. StandardScaler and MinMaxScaler are compared using training rows. The persisted Joblib transformer is reusable for compatible records and ignores categories unseen during fit.

The output is a preprocessing artifact, not a trained model. In the next task, fit this preprocessing inside the model-validation pipeline so that each validation fold learns imputation, capping, encoding, and scaling only from its training fold. Revisit clipping/scaling choices for the selected estimator.

Dataset attribution and references

IBM HR Analytics Employee Attrition & Performance dataset listing: Kaggle (fictional IBM sample), https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset. A public GitHub mirror provided the CSV used for this reproducible project. Analysis libraries: pandas, NumPy, scikit-learn, Matplotlib, Seaborn, and Joblib.